

İZMİR DEMOKRASİ UNIVERSITY



MIS203 / WEEK 02

Variables, Types And Expressions

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This Week In Python

- 1 Convert text input to numbers
- 2 Use arithmetic with clear parentheses
- 3 Format a small business quote

**A customer wants a two-item quote.
Which numbers must the program
calculate?**

What would the program need to know?

Choose The Right Value Type

The same characters can mean different things.

1. "25" is text. 25 is a whole number.
2. 25.5 is a decimal number.
3. Convert input before a calculation.

Example: `examples/01_types_and_conversion.py`

Turn Input Into Numbers

Conversion makes arithmetic possible.

```
quantity = int(input("Quantity: "))  
price = float(input("Unit price: "))  
subtotal = quantity * price  
print(f"Subtotal: {subtotal:.2f} TRY")
```

Example: $3 \times 125.50 \text{ TRY} = 376.50 \text{ TRY}$.

Arithmetic Operators In Python

The same numbers show how each symbol behaves.

+	Addition	$7 + 2 = 9$
-	Subtraction	$7 - 2 = 5$
*	Multiplication	$7 * 2 = 14$
/	Division	$7 / 2 = 3.5$
//	Floor Division	$7 // 2 = 3$
%	Remainder	$7 \% 2 = 1$
**	Exponentiation	$7 ** 2 = 49$

Example file: `examples/02_operators.py`

Revenue, Profit And Profit Margin

Profit margin is the percentage of revenue left after costs.

1. Revenue = quantity $\times$ unit price.
2. Profit = revenue – cost.
3. Profit margin (%) = profit $\div$ revenue $\times$ 100.

Example: 200 TRY revenue, 150 TRY cost, 25% profit margin.

Parentheses Clarify Precedence

These expressions do different things.

```
price = 120  
quantity = 3  
delivery = 40  
print(price * quantity + delivery)  
print(price * (quantity + delivery))
```

Which line represents a real purchase total?

Make Output Easy To Check

A number without a label is hard to interpret.

1. Show what was calculated.
2. Show the unit, such as TRY or percent.
3. Use two decimal places for a quote.

Example: `examples/06_format_a_quote.py`

Build An Invoice Line

Separate intermediate results for easier debugging.

```
quantity = 2
unit_price = 50.0
subtotal = quantity * unit_price
tax = subtotal * 0.10
print(f"Total: {subtotal + tax:.2f} TRY")
```

Change the tax rate and predict the result.

Find The Calculation Error

This program runs but prints the wrong total.

```
price = 50  
quantity = 2  
delivery = 20  
total = price * (quantity + delivery)  
print(total)
```

Fix the expression and explain the change.

Live Coding: Purchase Quote

Build a two-item quote and test it.

1. Read quantities and prices from input.
2. Calculate each line and a subtotal.
3. Add tax and delivery, then print a readable total.

A small test: $2 \times 50 + 1 \times 80 + 20$ delivery, 10% tax on 180 = 218 TRY.

Project A: Invoice Estimator

Compute two item lines, discount, tax and shipping.

`projects/project_a_invoice_estimator.py`

Trace each intermediate amount.

Project B: Break-Even Explorer

Estimate contribution margin and the required unit count.

`projects/project_b_break_even.py`

Example edge case: unit price below variable cost.

In-Class Lab: Two-Item Purchase Quote

Your Challenge

Assignment: `week02/lab-quiz/lab_assignment.md`

Submit in your repository: `week02/lab-quiz/`

Build it. Check a boundary. Explain one decision.

GitHub Checkpoint

Lab and quiz: week02/lab-quiz/

Keep weekly homework in week02/ outside that folder.

```
git add week02/lab-quiz/ README.md  
git commit -m "week02: complete lab quiz"  
git push
```

Show the lab, one test and the commit.

See You Next Week